

Quiz 4

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find a general solution using the substitution method for the differential equation:

$$y' = (9x + y)^2$$

(Hint: using the substitution $v = 9x + y$)

let $V = 9x + y$, then $y = V - 9x$ and

$$\frac{dy}{dx} = \frac{dV}{dx} - 9$$

Thus we have

$$\frac{dV}{dx} - 9 = V^2$$

$$\frac{dV}{dx} = V^2 + 9$$

$$\int \frac{1}{V^2 + 9} dV = \int dx$$

$$\frac{1}{3} \tan^{-1}\left(\frac{V}{3}\right) = x + C_1 \quad (C = 3C_1)$$

$$\frac{V}{3} = \tan(3x + C)$$

$$V = 3 \tan(3x + C)$$

$$\Rightarrow y = 3 \tan(3x + C) - 9x$$

Problem 2. (5 points) Find a general solution using the substitution method for the differential equation:

$$xy^2y' = 3x^3 + y^3.$$

(Hint: using the substitution $v = y/x$)

$$y' = 3\frac{x^2}{y^2} + \frac{y}{x}$$

$$\text{let } v = \frac{y}{x}, \text{ then } y = vx \text{ and } \frac{dy}{dx} = v + x\frac{dv}{dx}$$

thus we have

$$x + x\frac{dv}{dx} = \frac{3}{v^2} + x$$

$$x\frac{dv}{dx} = \frac{3}{v^2}$$

$$\int v^2 dv = \int \frac{3}{x} dx$$

$$\Rightarrow \frac{v^3}{3} = 3\ln|x| + C_1$$

$$v^3 = 9\ln|x| + C \quad (C = 3C_1)$$

$$\Rightarrow \left(\frac{y}{x}\right)^3 = 9\ln|x| + C$$

$$y^3 = x^3(9\ln|x| + C)$$

$$y = x(9\ln|x| + C)^{\frac{1}{3}}$$